

GDU-net: A cross-gated dual-domain optimization unfolding network for image compressive sensing

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HIGHLIGHTS

- Cross-domain consistency motivates update-specific gated state communication.
- Step-specific gates realize prior injection, interaction, and state feedback.
- Image-wavelet reconstruction is unfolded as directional dual-domain correction.
- A learnable tree prior captures adaptive multi-scale wavelet dependencies.
- Uncertainty-guided refinement reduces the influence of unreliable cross-domain discrepancies.

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ABSTRACT

Deep unfolding networks (DUNs) have advanced image compressive sensing (CS) by embedding iterative optimization into learnable reconstruction models. However, many existing DUNs retain a single dominant reconstruction representation, while transform-domain information is used mainly as an auxiliary prior or fused feature. Moreover, conventional dual-domain fusion does not explicitly specify how cross-domain information should participate in individual solver steps, limiting the coordination of image-domain structures and wavelet-domain details. To address this limitation, we propose GDU-Net, a cross-gated dual-domain unfolding network that formulates image- and wavelet-domain representations as coupled reconstruction variables. Based on their dependencies and update order in the unfolded optimizer, three learnable gates perform wavelet-prior injection before image-domain descent, cross-domain conditioning during proximal denoising, and image-to-wavelet feedback after image-domain refinement. Furthermore, an adaptive tree-structured module is developed to replace fixed wavelet-tree shrinkage with learnable, content-adaptive threshold modulation. By explicitly modeling parent-child dependencies, it introduces structure-aware multi-scale wavelet priors into reconstruction. A cross-domain uncertainty-aware mechanism is further incorporated to predict spatial uncertainty maps and reduce the propagation of unreliable intermediate information. Extensive experiments demonstrate that GDU-Net achieves competitive reconstruction performance across benchmark datasets and shows improved robustness under challenging sampling and noise conditions.

1. Introduction

Compressive Sensing (CS) [12] exploits the inherent sparsity or compressibility of signals $\mathbf{x}$ to enable accurate reconstruction from limited

measurements. It comprises two fundamental processes: sampling and reconstruction. The sampling process is modeled as a linear transformation $\mathbf{y} = \mathbf{Ax}$ [75]. The reconstruction process requires solving an ill-posed inverse problem, typically formulated as a sparsity-constrained

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optimization problem:

$$\min_{\mathbf{x}} \|\mathbf{x}\|_1 \quad \text{s.t.} \quad \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2 \leq \delta. \quad (1)$$

Here, $\mathbf{A} \in \mathbb{R}^{M \times N}$ denotes the CS sampling matrix, and $\frac{M}{N}$ ($M \ll N$) indicates the CS sampling ratio. $\mathbf{y} \in \mathbb{R}^M$ represents the CS measurements. $\|\cdot\|_1$ and $\|\cdot\|_2$ denote the ℓ_1 and ℓ_2 norms, respectively. δ is a tiny positive parameter.

The significant challenge in CS reconstruction stems directly from its ill-posed nature and the vastness of its feasible solution space. Therefore, it is essential to constrain the solution space by incorporating powerful prior knowledge. Traditional model-based methods achieve this by utilizing intrinsic signal structures, including low-rank [11,19], non-local [41,71,76,79], and sparsity in certain transformation domains [1,77]. Among them, the wavelet tree-structured prior has gained significant attention due to its ability to capture the hierarchical dependencies of wavelet coefficients. Specifically, wavelet coefficients exhibit a quad-tree structure where significant coefficients at coarse scales (parent nodes) tend to propagate to finer scales (child nodes). The wavelet tree-structured prior can promote sparsity not only at the individual coefficient level but also across groups of coefficients, effectively capturing the inter-scale correlations inherent in images. Moreover, some nonlinear iterative methods have been widely employed in signal reconstruction, such as matching pursuit and orthogonal matching pursuit [37,55], approximate message passing (AMP) [13], iterative shrinkage thresholding (ISTA) [2], alternating direction method of multipliers (ADMM) [3], and so on. Although these methods offer significant interpretability and convergence guarantees, they still suffer from high computational complexity and parameter-selection challenges.

Recently, deep learning-based methods have achieved significant success. Pure deep learning-based methods [15,42,54] directly reconstruct signals from a small amount of CS measurements through data-driven learning. Such methods commonly utilize a series of non-linear modules to extract and refine deep features, including convolutional neural networks (CNNs) [46], Transformer [7,45,51], generative models [28], and so on. However, these pure deep learning-based methods are often trained as black boxes, making their decision processes difficult to explain. Thus, deep unfolding networks (DUNs) [30,31,85] are proposed, which can combine the interpretability of iterative algorithms with the efficiency of deep networks. DUNs offer superior performance by unfolding well-known iterative algorithms, such as ISTA-Net [73], AMP-Net [38,80], ADMM-CSNet [62], USB-Net [21], and PCNet [4]. Many studies have jointly improved the sampling and reconstruction processes of image CS. MCFD-Net [48] combines dynamic multi-cross sampling with DWT-based high- and low-frequency reconstruction. RHT-Net [47] combines representation-domain sampling with global and local attention in an unfolding network. To better preserve sharp edges and fine details, many unfolding-based CS methods [6,8,82] also introduce edge-aware or gradient-domain priors, often drawing inspiration from total variation. This trend aligns with the broader evolution of CS, as surveyed in recent works on physics-inspired deep unfolding [72] and advances in image/video CS [61,81].

However, many existing DUNs still perform reconstruction within a single dominant representation, which limits their ability to simultaneously preserve image-domain structural continuity and transform-domain high-frequency details. To overcome this limitation, recent studies have begun exploring information from different domains [23,59] and data uncertainty [56,83] to further improve performance. *Although these methods introduce complementary representations, cross-domain interaction is often implemented as feature-level fusion, such as concatenation, addition, or global attention-based weighting. Such designs enhance representation diversity, but do not explicitly specify how cross-domain information should participate in different optimization steps of an unfolded solver.* Consequently, cross-domain cues often remain generic fused features rather than serving step-specific optimization roles for correcting reconstruction errors during iterative updates. Meanwhile, most uncertainty

modeling works [14,56] focus on confidence estimation within a single domain, failing to explicitly model and leverage the discrepancy in uncertainty between the spatial and frequency domains to guide reconstruction. Moreover, the reliability and complementary roles of image- and wavelet-domain representations may vary across unfolding stages and sampling conditions. Fixed or generic feature-level fusion does not explicitly account for such variations, motivating adaptive cross-domain communication along the unfolding trajectory. In this work, we revisit dual-domain CS reconstruction from the perspective of optimization consistency. We formulate image- and wavelet-domain variables as a coupled optimization framework and unfold their interaction into three gated communication steps, enabling prior injection, dynamic interaction, and state feedback rather than generic feature-level fusion. The main contributions are summarized as follows:

- We propose a cross-gated dual-domain unfolding framework to address the lack of step-specific interaction in conventional dual-domain fusion. Following the alternating update order, three update-specific gates perform wavelet-prior-guided image updating, dual-domain proximal conditioning, and image-to-wavelet feedback. This design enables adaptive and directional cross-domain communication throughout the unfolding process.
- We develop an adaptive tree-structured wavelet module (ATSM) to address the limited adaptability of handcrafted wavelet shrinkage and coefficient-wise sparsity priors. This ATSM replaces the fixed group-wise shrinkage with a learnable, structure-aware thresholding surrogate. It models parent-child dependencies and adaptively modulates cross-scale wavelet coefficients.
- We introduce a cross-domain uncertainty-aware mechanism to limit the propagation of unreliable information. The epistemic-style cross-domain discrepancy map guides image-domain refinement, while its combination with the aleatoric uncertainty map reweights supervision at multiple unfolding stages.

2. Related work

2.1. Single-domain methods

Single-domain methods have been widely studied in the field of CS. [39] first introduced the stacked autoencoder as an effective technique for reconstructing images from CS measurements. [29] proposed ReconNet, a classical CNN-based framework for CS reconstruction. [64] proposed DR²-Net to integrate residual learning within its CNN-based model. [62] proposed ADMM-CSNet, an unfolding network of the ADMM algorithm. [73] transformed the well-known ISTA algorithm into a deep network named ISTA-Net. [46] developed CSNet to optimize both the sampling and reconstruction processes jointly. From the perspective of denoising, AMP-Net [80] unrolled each iteration of the AMP algorithm into a corresponding network module. [66] proposed CSformer, which seamlessly combines the Transformer with CNN-based components. PCNet [4] is a lightweight deep CS network with an efficient design. Despite their great success, these methods rely solely on single-domain information, limiting their ability to fully exploit cross-domain correlations and preserve global structural consistency.

2.2. Dual-domain and dual-path methods

To overcome the limitations of single-domain approaches, dual-domain frameworks have emerged as a robust alternative. [43] proposed a parallel dual-domain CNN framework for CS-MRI, utilizing the correlation between k -space and the spatial domain. [25] developed HDNet to merge pixel-level focus with frequency-domain details. [60] proposed a dual-domain residual-based optimization framework for sparse-view CT reconstruction. [84] designed DuDoRNet to utilize a deep prior and a dilated residual dense network for MRI reconstruction. [9] introduced a dual-domain attention mechanism to bridge the gap between spatial and frequency domains. [67] enhanced data representation by integrating features from dual domains. [59] designed a novel framework

to exploit dual-domain features. Furthermore, dual-path structures in DUNs [16,20,44] enhanced performance through dedicated pathways for feature flow or multi-scale learning. DDFNet [33], FredNet [34], and FSSFFormer [32] exploit spatial–frequency fusion, frequency-guided denoising, and wavelet-based spectral–spatial modeling, respectively. While these methods show promising performance, most rely on static feature-level fusion (e.g., concatenation) or adaptive weighting, lacking explicit correspondence to optimization steps. Their goal is to enhance feature representation, whereas ours orchestrates controllable, stepwise information flow between domains via optimization-inspired interaction. Our cross-domain gates are aligned with specific steps of the unfolded optimizer and provide optimization-informed cross-domain guidance in a learnable manner.

2.3. Uncertainty modeling in inverse problems

Uncertainty modeling has gained increasing attention in image recovery tasks. [65] developed a network that used uncertainty to guide the multi-scale learning of rainfall patterns. [53] introduced a variational deep probabilistic model to quantify the uncertainty in image reconstruction. [14] employed variational autoencoders (VAEs) to encode acquisition uncertainty in a latent code. [40] proposed an adaptive weighted loss function to reduce the high uncertainty associated with complex image structures and textures. [36] introduced uncertainty-aware regularization into the GAN network to enhance its robustness. Despite these advances, existing methods often fail to explicitly quantify uncertainty across different domains.

3. Proposed method

3.1. Dual-domain unfolding architecture

As shown in Fig. 1, our proposed framework utilizes the complementary features from both the image and wavelet domains, enabling a more comprehensive representation of various image components. Our framework consists of three key parts: sampling, initialization, and reconstruction.

3.1.1. Sampling

The sampling process is used to acquire CS measurements from the original signal. For CS sampling, the entire image $X \in \mathbb{R}^{H \times W}$ is first divided into blocks x_i of size 64×64 . After reshaping one image block

into a vector, the sampling process can be performed on such vectors, which can be formulated as:

$$y_i = A x_i, \quad (2)$$

where $x_i \in \mathbb{R}^N$ ($N = 64 \times 64$) is the i -th image block and $y_i \in \mathbb{R}^M$ represents the CS measurements. $A \in \mathbb{R}^{M \times N}$ indicates the block-based sampling matrix.

3.1.2. Initialization

The initialization process produces an initial restored image by employing a linear transformation, which can be formulated as follows:

$$x_i^{(0)} = A^T y_i, \quad (3)$$

$$\theta_i^{(0)} = D(x_i^{(0)}), \quad (4)$$

where $A^T \in \mathbb{R}^{N \times M}$ denotes the transpose of the CS sampling matrix and $x_i^{(0)} \in \mathbb{R}^N$ is the initialization of an image block. The deep reconstruction builds upon this initial reconstruction.

3.1.3. Dual-domain reconstruction

The reconstruction process unfolds over K stages, each corresponding to an iteration of the optimization algorithm. The proposed dual-domain model is formulated as follows:

$$\arg \min_{x, \theta} \frac{1}{2} \|Ax - y\|_2^2 + \frac{\alpha}{2} \|\theta - D(x)\|_2^2 + \lambda G(\theta) + \mu F(x), \quad (5)$$

where $D(\cdot)$ denotes the orthonormal Haar discrete wavelet transform (DWT), satisfying $D^T = D^{-1}$ and $D^T D = I$. $\|Ax - y\|_2^2$ indicates the data-fidelity term, while $\|\theta - D(x)\|_2^2$ enforces consistency between the image-domain variable and the wavelet-domain variable. $G(\theta)$ denotes the tree-structured prior within the wavelet domain, as shown in Fig. 3(a) and $F(x)$ is the regularization term within the image domain. A is the CS sampling matrix. θ denotes the reconstructed wavelet coefficients. x indicates the reconstructed image. The parameter α controls the cross-domain consistency strength and is instantiated as a stage-specific learnable coefficient $\alpha^{(k)}$ in the unfolded network. The regularization weights λ and μ are introduced at the model level but are not retained as independently optimized scalar parameters in the practical implementation. Specifically, the wavelet-domain regularization associated with λ

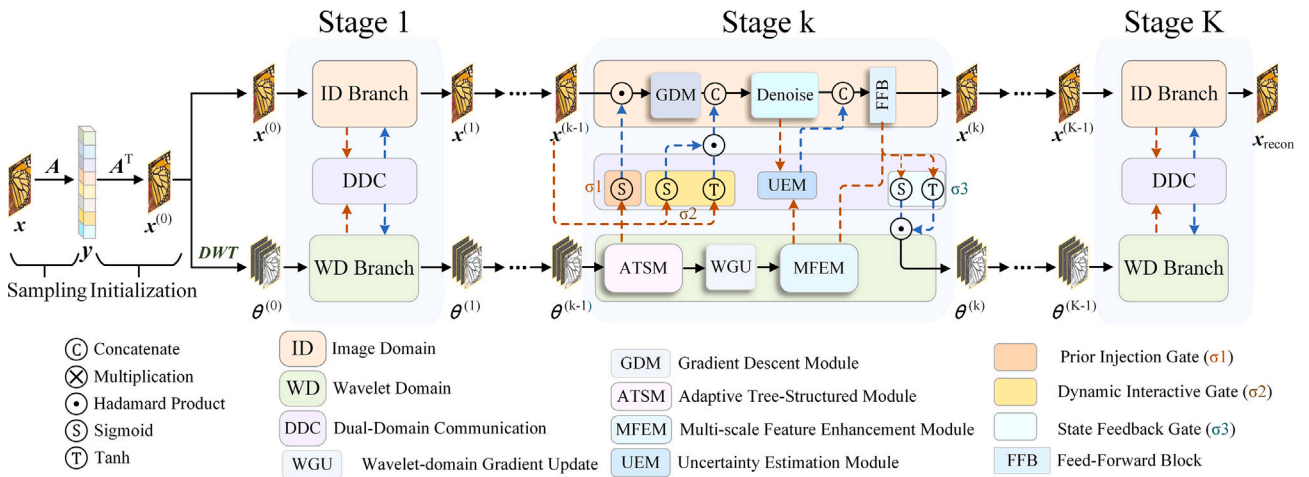


Fig. 1. Overall architecture of the proposed GDU-Net, consisting of K unfolding stages with coupled image- and wavelet-domain branches. DDC denotes dual-domain communication, while WD and ID denote the wavelet and image domains, respectively. Solid arrows represent the main optimization trajectory, whereas dashed arrows denote step-specific cross-domain communication. GDM and WGU perform the image- and wavelet-domain gradient updates, respectively. ATSM produces the prior-informed auxiliary wavelet state z , which is used by WGU to obtain the preliminary wavelet variable w before learned multiscale refinement by MFEM. UEM estimates cross-domain reconstruction uncertainty, and FFB performs uncertainty-aware image-domain refinement. The three gates, namely the prior-injection, dynamic-interaction, and state-feedback gates, regulate directional information transfer at their corresponding optimization steps. The detailed GDM and WGU updates are given in Eqs. (7) and (15), respectively.

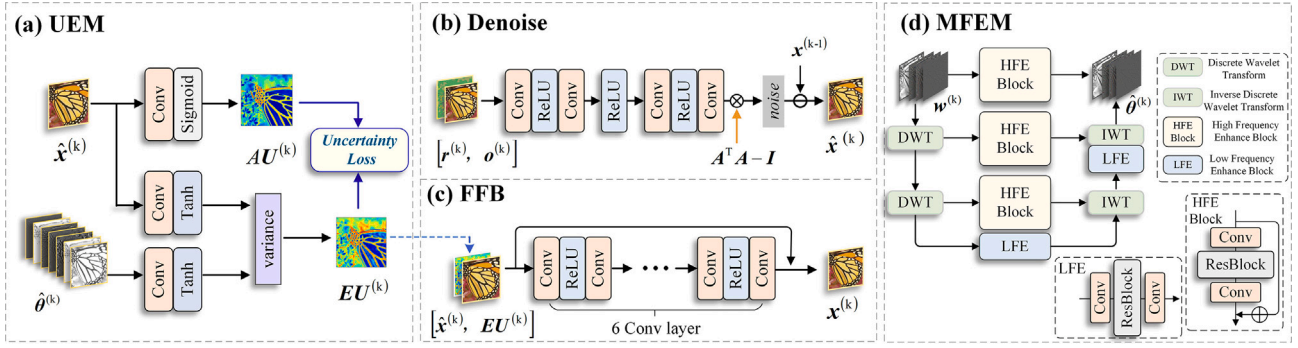


Fig. 2. Detailed structures and information flow of the functional modules in GDU-Net. UEM estimates the reliability of the intermediate dual-domain representations and provides uncertainty guidance for subsequent refinement. The Denoise module performs learned proximal reconstruction. FFB uses the estimated cross-domain uncertainty to perform uncertainty-aware image-domain refinement, suppress unreliable responses, and produce the final image-domain output of the current stage. MFEM employs a three-level DWT/IWT encoder-decoder architecture. At each level, HFE refines the concatenated HL, LH, and HH components, while LFE processes the deepest LL component before progressive IWT reconstruction.

is parameterized by ATSM and MFEM, whereas the image-domain regularization associated with μ is represented by the learned denoising module.

To simplify the solution process, we decompose the problem represented by Eq. (5) into two subproblems, each associated with a distinct branch of our framework.

Image-Domain Branch: The decoupled image-domain subproblem is formulated as follows:

$$\arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \frac{\alpha}{2} \|\boldsymbol{\theta} - D(\mathbf{x})\|_2^2 + \mu F(\mathbf{x}). \quad (6)$$

We use the following steps to solve the model in Eq. (6).

$$\mathbf{r}^{(k+1)} = \mathbf{x}^{(k)} - \eta^{(k)} (\mathbf{A}^T (\mathbf{A}\mathbf{x}^{(k)} - \mathbf{y}) + \alpha^{(k)} (\mathbf{x}^{(k)} - \mathbf{x}_{\theta}^{(k)})), \quad (7)$$

$$\hat{\mathbf{x}}^{(k+1)} = \text{Denoise}(\mathbf{r}^{(k+1)}), \quad (8)$$

where $\text{Denoise}(\cdot)$ denotes the denoising sub-network, corresponding to Fig. 2(b). $\mathbf{x}_{\theta}^{(k)} = D^{-1}(\boldsymbol{\theta}^{(k)})$, D^{-1} denotes the inverse DWT, and $\eta^{(k)}$ denotes the stage-specific learnable step size of the image-domain gradient update.

Wavelet-Domain Branch: The decoupled wavelet-domain subproblem is formulated as follows:

$$\arg \min_{\boldsymbol{\theta}} \frac{\alpha}{2} \|\boldsymbol{\theta} - D(\mathbf{x})\|_2^2 + \lambda G(\boldsymbol{\theta}). \quad (9)$$

To effectively utilize the hierarchical nature of images, we employ a tree-structured prior (TSP) that organizes wavelet coefficients in a quadtree structure [5]. This prior characterizes the parent-child dependency of wavelet coefficients across scales and encourages structured sparsity in the wavelet domain. Specifically, the wavelet-domain regularizer is defined as

$$G(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|_1 + \sum_{p \in \mathcal{P}} \|\tilde{\boldsymbol{\theta}}_p\|_2, \quad (10)$$

where $\mathcal{P}$ denotes the set of all parent-child groups, and p indexes one group in $\mathcal{P}$. $\tilde{\boldsymbol{\theta}}_p$ denotes the extended coefficient vector associated with group p . The first term enforces coefficient-level sparsity, while the second term promotes group-level sparsity by summing the ℓ_2 norms of hierarchical parent-child groups. Thus, the prior jointly models individual sparsity and cross-scale structural dependency in the wavelet domain.

The model in Eq. (9) can therefore be written as

$$\arg \min_{\boldsymbol{\theta}} \frac{\alpha}{2} \|\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{im}}\|_2^2 + \lambda \left(\|\boldsymbol{\theta}\|_1 + \sum_{p \in \mathcal{P}} \|\tilde{\boldsymbol{\theta}}_p\|_2 \right). \quad (11)$$

where $\boldsymbol{\theta}_{\text{im}} = D(\mathbf{x})$ is obtained by wavelet decomposition of the image-domain features.

To avoid ambiguity, we use $\mathcal{P}$ only for the group set, and denote by $\mathbf{G}_p$ the operator that extracts the grouped parent-child coefficients of group p from $\boldsymbol{\theta}$. Let $\mathbf{G}$ be the stacked grouping operator composed of all $\{\mathbf{G}_p\}_{p \in \mathcal{P}}$, and introduce an auxiliary grouped variable $\mathbf{z} = \{\mathbf{z}_p\}_{p \in \mathcal{P}}$ with the constraint $\mathbf{z} = \mathbf{G}\boldsymbol{\theta}$, i.e., $\mathbf{z}_p = \mathbf{G}_p\boldsymbol{\theta}$. We relax this auxiliary equality constraint by introducing a quadratic penalty term, yielding the following penalized problem:

$$\arg \min_{\boldsymbol{\theta}} \frac{\alpha}{2} \|\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{im}}\|_2^2 + \lambda (\|\boldsymbol{\theta}\|_1 + \|\mathbf{z}\|_{2,1}) + \frac{\gamma}{2} \|\mathbf{z} - \mathbf{G}\boldsymbol{\theta}\|_2^2, \quad (12)$$

where $\|\mathbf{z}\|_{2,1} = \sum_{p \in \mathcal{P}} \|\mathbf{z}_p\|_2$ denotes the group sparsity penalty. γ is introduced as the quadratic-penalty coefficient for the auxiliary constraint $\mathbf{z} = \mathbf{G}\boldsymbol{\theta}$. In the corresponding model-based solver, the ratio λ/γ determines the fixed group-shrinkage threshold in Eq. (14). In the practical unfolded network, this fixed threshold λ/γ is replaced by the content-adaptive threshold predicted by ATSM. Meanwhile, the coefficient-wise proximal mapping associated with $\lambda\|\boldsymbol{\theta}\|_1$ is parameterized by MFEM. Therefore, λ and γ are not retained as independently optimized scalar parameters.

Let

$$f(\boldsymbol{\theta}) = \frac{\alpha}{2} \|\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{im}}\|_2^2 + \frac{\gamma}{2} \|\mathbf{z} - \mathbf{G}\boldsymbol{\theta}\|_2^2. \quad (13)$$

We solve the wavelet-domain model through the following iterative steps:

$$\mathbf{z}_p^{(k+1)} = \frac{\mathbf{G}_p\boldsymbol{\theta}^{(k)}}{\|\mathbf{G}_p\boldsymbol{\theta}^{(k)}\|_2 + \epsilon} \max\left(\|\mathbf{G}_p\boldsymbol{\theta}^{(k)}\|_2 - \frac{\lambda}{\gamma}, 0\right), \quad p \in \mathcal{P}, \quad (14)$$

$$\mathbf{w}^{(k+1)} = \boldsymbol{\theta}^{(k)} - \tau^{(k)} \nabla f(\boldsymbol{\theta}^{(k)}), \quad (15)$$

$$\hat{\boldsymbol{\theta}}^{(k+1)} = \text{MFEM}(\mathbf{w}^{(k+1)}), \quad (16)$$

where ϵ is a small positive constant for numerical stability, $\tau^{(k)}$ denotes the stage-specific learnable step size of the wavelet-domain gradient update, and $\mathbf{z}^{(k+1)} = \{\mathbf{z}_p^{(k+1)}\}_{p \in \mathcal{P}}$. When $\|\mathbf{G}_p\boldsymbol{\theta}^{(k)}\|_2 = 0$, the corresponding shrinkage output is defined as zero. The gradient is given by

$$\nabla f(\boldsymbol{\theta}^{(k)}) = \alpha^{(k)} (\boldsymbol{\theta}^{(k)} - \boldsymbol{\theta}_{\text{im}}^{(k)}) + \gamma \mathbf{G}^T (\mathbf{G}\boldsymbol{\theta}^{(k)} - \mathbf{z}^{(k+1)}). \quad (17)$$

Here, $\boldsymbol{\theta}_{\text{im}}^{(k)} = D(\mathbf{x}^{(k)})$. Eq. (14) implements group-wise wavelet-tree shrinkage under the TSP, which is further unfolded into the proposed adaptive tree-structured module described in Section 3.2. Instead of using fixed handcrafted thresholds, the unfolded module learns data-dependent hierarchical modulation for more effective sparsity modeling and texture preservation. Eq. (17) presents the model-based gradient form.

The subsequent MFEM(·) performs three-scale wavelet-domain refinement, as shown in Fig. 2. Its input $\mathbf{u}^{(k+1)} \in \mathbb{R}^{B \times 4 \times H/2 \times W/2}$ is the level-1 wavelet representation $\text{Concat}(\mathbf{LL}_1, \mathbf{LH}_1, \mathbf{HL}_1, \mathbf{HH}_1)$ obtained from the first DWT. Only the low-frequency components are recursively decomposed to construct Levels 2 and 3. At each level ℓ , the high-frequency subbands are concatenated as $\mathbf{H}_\ell = \text{Concat}(\mathbf{HL}_\ell, \mathbf{LH}_\ell, \mathbf{HH}_\ell)$ and processed by an HFE consisting of a $3 \rightarrow 32$ convolution, a 32-channel ResBlock, and a $32 \rightarrow 3$ output convolution, followed by a residual connection. The deepest $\mathbf{LL}_3$ is processed by an LFE with a $1 \rightarrow 32 \rightarrow 1$ structure. During decoding, IWT progressively combines the refined low-frequency feature with the enhanced high-frequency features at Levels 3 and 2. The reconstructed $\mathbf{LL}_1$ is finally concatenated with the enhanced Level-1 high-frequency subbands to produce $\hat{\theta}^{(k+1)}$ with the same size as $\mathbf{u}^{(k+1)}$. Thus, MFEM refines all four wavelet subbands while preserving a wavelet-domain output.

Dual-Domain Communication Gates: The coupled objective determines the explicit reconstruction variables, their bidirectional consistency dependencies, and the positions at which cross-domain information enters the alternating update. To realize the dual-domain coupling in Eq. (5), we introduce three learnable gating functions ($\sigma_1, \sigma_2, \sigma_3$) within the unfolded optimization framework. By using lightweight convolutional layers with Sigmoid/Tanh activations, these gates adaptively regulate cross-domain interactions and serve as learnable parameterizations inspired by iterative optimization updates.

Prior Injection Gate (σ_1): Consider the gradient descent step for the image-domain subproblem (Eq. 7). The term $\alpha^{(k)}(\mathbf{x}^{(k)} - \mathbf{x}_\theta)$ provides a fixed linear feedback from the wavelet domain to guide the image update. The coefficient α in the consistency gradient is retained. Rather than replacing α , σ_1 uses the prior-informed wavelet estimate $\mathbf{z}^{(k+1)}$ to condition a spatial mask for the image-domain state before the descent step. It should therefore be interpreted as a learned wavelet-prior-conditioned modulation, rather than an exact discretization of the consistency gradient. Specifically, we parameterize this feedback as:

$$\mathbf{x}_{\sigma_1}^{(k)} = \text{Sigmoid}(\mathbf{W}_{wi} * \mathbf{z}_z^{(k+1)} + \mathbf{W}_{xi} * \mathbf{x}^{(k)}) \odot \mathbf{x}^{(k)}. \quad (18)$$

Here, $\odot$ denotes element-wise multiplication for spatially adaptive feature modulation, following the notation used in Fig. 1. This gate σ_1 (shown in Fig. 1) effectively learns to reweight each pixel in $\mathbf{x}^{(k)}$ based on the joint information from both domains. Therefore, we further replace $\mathbf{x}^{(k)}$ with gated representation $\mathbf{x}_{\sigma_1}^{(k)}$, yielding the practical implementation:

$$\mathbf{r}^{(k+1)} = \mathbf{x}_{\sigma_1}^{(k)} - \eta^{(k)}(\mathbf{A}^\top(\mathbf{A}\mathbf{x}_{\sigma_1}^{(k)} - \mathbf{y}) + \alpha^{(k)}(\mathbf{x}_{\sigma_1}^{(k)} - \mathbf{x}_\theta^{(k)})). \quad (19)$$

Dynamic Interactive Gate (σ_2): Within the image-domain denoising step (Eq. 8), this gate conditions the denoiser on cross-domain context, which corresponds to the proximal operator for the regularizer $\mathcal{F}(\mathbf{x})$. It dynamically assesses the relevance between wavelet-domain sparse features $\mathbf{z}^{(k+1)}$ and image-domain features $\mathbf{x}^{(k)}$, producing a fused gating signal $\mathbf{o}^{(k+1)}$ that enables the denoising process to be aware of both spatial structures and wavelet-frequency features. This process corresponds to σ_2 of Fig. 1.

$$\mathbf{d}^{(k+1)} = \text{Sigmoid}(\mathbf{W}_{wd} * \mathbf{z}_z^{(k+1)} + \mathbf{W}_{xd} * \mathbf{x}^{(k)}), \quad (20)$$

$$\mathbf{j}^{(k+1)} = \text{Tanh}(\mathbf{W}_{wj} * \mathbf{z}_z^{(k+1)} + \mathbf{W}_{xj} * \mathbf{x}^{(k)}), \quad (21)$$

$$\mathbf{o}^{(k+1)} = \mathbf{d}^{(k+1)} \odot \mathbf{j}^{(k+1)}. \quad (22)$$

Here, $\mathbf{x}_z^{(k+1)} = D^{-1}(\mathcal{R}(\mathbf{z}^{(k+1)}))$ denotes the image-domain representation obtained by applying the inverse wavelet transform to $\mathbf{z}^{(k+1)}$. $\mathcal{R}(\cdot)$ restores the grouped coefficients to their original wavelet-domain positions. The denoising operation in Eq. (8) is therefore enhanced as:

$$\hat{\mathbf{x}}^{(k+1)} = \text{Denoise}(\mathbf{r}^{(k+1)}, \mathbf{o}^{(k+1)}). \quad (23)$$

As depicted in Fig. 2(c), the Feed-Forward Block (FFB) refines the denoised image-domain estimate under the guidance of the cross-domain

uncertainty map, thereby suppressing unreliable responses and producing the final image-domain state of the current stage. Meanwhile, $\mathbf{EU}^{(k+1)}$ is obtained from the Uncertainty Estimation Module (UEM), which is described in Section 3.3.

$$\mathbf{x}^{(k+1)} = \text{FFB}(\hat{\mathbf{x}}^{(k+1)}, \mathbf{EU}^{(k+1)}). \quad (24)$$

The placement of σ_2 at the denoising step is motivated by the proximal update, whereas the cross-domain condition and its Sigmoid/Tanh implementation are learned extensions rather than an exact proximal operator derived from $\mathcal{F}$.

State Feedback Gate (σ_3): This gate implements the feedback step of the alternating optimization between the image-domain and wavelet-domain subproblems (Eqs. (6) and (9)). It performs a learned, adaptive fusion of the latest image-domain reconstruction $\mathbf{x}^{(k+1)}$ with the preliminary wavelet-domain estimate $\hat{\theta}^{(k+1)}$, thereby providing a learned consistency-inspired feedback from the latest image estimate to the wavelet state $\theta \approx D(\mathbf{x})$ in Eq. (5). This process corresponds to σ_3 of Fig. 1, and is formulated as:

$$\mathbf{s}^{(k+1)} = \text{Sigmoid}(\mathbf{W}_{ws} * \mathbf{x}_{\theta}^{(k+1)} + \mathbf{W}_{xs} * \mathbf{x}^{(k+1)}), \quad (25)$$

$$\mathbf{f}^{(k+1)} = \text{Tanh}(\mathbf{W}_{wf} * \mathbf{x}_{\theta}^{(k+1)} + \mathbf{W}_{xf} * \mathbf{x}^{(k+1)}). \quad (26)$$

Here, $\mathbf{x}_{\theta}^{(k+1)} = D^{-1}(\hat{\theta}^{(k+1)})$ denotes the image-domain representation obtained by applying the inverse wavelet transform to $\hat{\theta}^{(k+1)}$. Following this gate, the $k+1$ -th reconstruction from the wavelet domain can be expressed as:

$$\theta^{(k+1)} = D(\text{Conv}_{1 \times 1}(\mathbf{s}^{(k+1)} \odot \mathbf{f}^{(k+1)})). \quad (27)$$

Here, $\mathbf{W}_{wi}, \mathbf{W}_{xi}, \mathbf{W}_{wd}, \mathbf{W}_{xd}, \mathbf{W}_{wj}, \mathbf{W}_{xj}, \mathbf{W}_{ws}, \mathbf{W}_{xs}, \mathbf{W}_{wf},$ and $\mathbf{W}_{xf}$ represent the learnable filter weights. $\odot$ denotes the Hadamard product. The three gates employ standard convolutional and nonlinear operations and are not proposed as new generic attention operators. Their role is to regulate the participation of one explicit reconstruction state in a prescribed update of the other while retaining separate image- and wavelet-domain states. The proposed design instantiates wavelet-to-image prior injection, dual-domain proximal conditioning, and image-to-wavelet feedback at distinct positions of each unfolded stage. Table 1 summarizes the transmission direction, optimization site, corresponding equations, and practical role of each gate.

3.2. Adaptive tree-structured module (ATSM)

For image restoration, multi-scale information is crucial for capturing hierarchical structures [27,78]. However, these methods typically operate in the spatial domain with fixed hierarchical structures (e.g., spatial pyramids), failing to fully exploit the natural multi-scale organization of wavelet transforms. Similarly, traditional wavelet tree priors rely on handcrafted, fixed thresholds. To overcome these limitations, we parameterize the proximal mapping induced by the tree-structured prior with a learnable adaptive thresholding mechanism, named ATSM, reformulating Eq. (14) through Eqs. (28)–(32). As shown in Fig. 3(b), the ATSM explicitly encodes the wavelet quadtree hierarchy via grouped parent-child and sibling dependencies, and adjusts thresholds in a data-driven manner, enabling adaptive learning of cross-scale dependencies directly from data.

First, the Group(·) operator aligns parent-child wavelet coefficient groups across scales with channels, thereby mapping parent-child relationships to channel-wise correlations through the following steps:

$$\theta_g = \text{Group}(\text{Divide}(\theta)). \quad (28)$$

Then, the grouped wavelet coefficients are fed into a spatial attention (SA) module, where a spatially adaptive threshold is predicted in

Table 1

Correspondence between the gated communication operators and the unfolded optimization steps.

Gate	Direction	Optimization Site	Practical Equations	Function
σ_1	WD $\rightarrow$ ID	Consistency feedback in the image-gradient step	Eqs. (18)–(19)	Spatially modulates the wavelet-informed image state before descent.
σ_2	WD + ID $\rightarrow$ ID	Image proximal/denoising step	Eqs. (20)–(23)	Conditions the denoiser on cross-domain context.
σ_3	ID $\rightarrow$ WD	Post-refinement consistency feedback	Eqs. (25)–(27)	Updates the wavelet state using the latest image estimate.

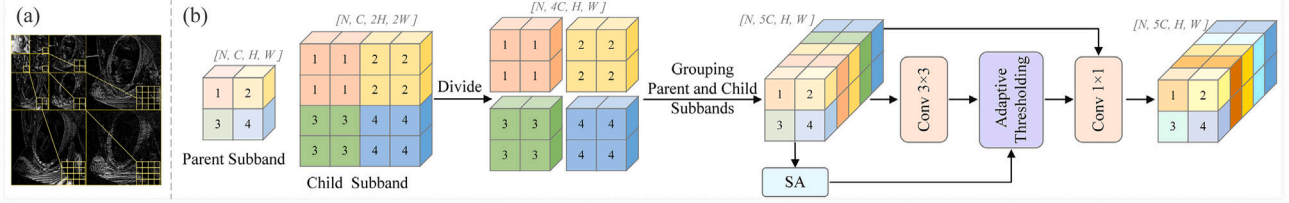


Fig. 3. Detailed structure of the adaptive tree-structured module (ATSM), as elaborated in Section 3.2. (a) Wavelet quadtree constructed from multi-scale wavelet coefficients, where each parent coefficient is associated with its spatially corresponding child coefficients at the adjacent finer scale. (b) Structure-aware coefficient processing in ATSM. The coefficients are rearranged and grouped according to their parent-child and sibling dependencies, after which spatial attention (SA) estimates content-adaptive threshold modulation. Adaptive thresholding is then applied to suppress unreliable coefficients while preserving structurally correlated details. SA is described in Eq. (29). The adaptive-thresholding process is outlined in Eq. (30).

a data-driven manner. The SA adapts thresholds based on the spatial characteristics:

$$\mathbf{V}_{\text{thr}} = \text{Sigmoid}(\text{Conv}_{3 \times 3}(\text{Concat}(\text{AvgPool}(\theta_g), \text{MaxPool}(\theta_g)))), \quad (29)$$

$$\tilde{\theta}_g = \frac{\theta_g}{\|\theta_g\|_2 + \epsilon} \max(\|\theta_g\|_2 - \mathbf{V}_{\text{thr}}, 0), \quad (30)$$

$$\mathbf{z} = \text{Conv}_{1 \times 1}(\text{Concat}(\theta_g, \tilde{\theta}_g)), \quad (31)$$

$$\mathbf{z}_r = \mathcal{R}(\mathbf{z}). \quad (32)$$

Here, $\theta_g \in \mathbb{R}^{B \times C_g \times H_g \times W_g}$. In Eq. (29), channel-wise average and maximum pooling are followed by convolution and a Sigmoid activation to produce $\mathbf{V}_{\text{thr}} \in \mathbb{R}^{B \times 1 \times H_g \times W_g}$. In Eq. (30), the ℓ_2 norm is computed along C_g at each spatial location, and $\mathbf{V}_{\text{thr}}$ is broadcast along C_g for spatially adaptive group-wise shrinkage. The operator $\mathcal{R}(\cdot)$ then restores the grouped coefficients to their original wavelet-domain positions. Compared with the fixed-threshold group-wise shrinkage in Eq. (14), ATSM preserves the parent-child grouping structure while replacing the handcrafted threshold λ/γ with a learnable, spatially adaptive threshold. Therefore, ATSM parameterizes the group shrinkage effect associated with λ/γ and produces the prior-informed auxiliary state before the wavelet-domain gradient update. MFEM subsequently serves as a learned proximal refinement for the preliminary wavelet-domain variable.

3.3. Uncertainty estimation module (UEM)

Since spatial and wavelet domains respond differently to structures and errors, their inconsistency reflects reconstruction reliability. Therefore, we estimate cross-domain inconsistency to guide reliability-aware optimization and suppress error accumulation.

To estimate the epistemic-style cross-domain discrepancy uncertainty (EU), which characterizes representation inconsistency between spatial and wavelet domains, the dual-domain features are projected into a latent space via a shared $\text{Conv}_{1 \times 1}$ layer with Tanh activation, producing calibrated predictions $\mathbf{u}_\theta^{(k)}, \mathbf{u}_x^{(k)}$.

$$\mathbf{u}_\theta^{(k)} = \text{Tanh}(\text{Conv}_{1 \times 1}(D^{-1}(\hat{\theta}^{(k)}))), \quad (33)$$

$$\mathbf{u}_x^{(k)} = \text{Tanh}(\text{Conv}_{1 \times 1}(\hat{\mathbf{x}}^{(k)})), \quad (34)$$

$$\mathbf{EU}^{(k)} = \text{variance}(\mathbf{u}_\theta^{(k)}, \mathbf{u}_x^{(k)}), \quad (35)$$

where $\text{variance}(\cdot)$ measures cross-domain representation discrepancy, producing an epistemic-style cross-domain discrepancy-based uncertainty map $\mathbf{EU}^{(k)}$ for unreliable-region awareness.

For aleatoric uncertainty, let $\mathbf{x}^{(k-1)}$ and $\mathbf{x}^{(k)}$ denote the input and output of the k -th stage, respectively, and let $\mathbf{x}^{\text{gt}}$ represent the ground-truth image. We assume that the reconstruction residual follows a heteroscedastic Gaussian distribution, whose variance is spatially adaptive and input-dependent. To model such uncertainty, a $\text{Conv}_{3 \times 3}$ layer is used to predict a log-variance map from the intermediate reconstruction. Its Sigmoid-normalized form is further used as an aleatoric uncertainty map for uncertainty-aware residual weighting.

At the k -th stage, the predicted log-variance map is given by

$$\mathbf{v}^{(k)} = \text{Conv}_{3 \times 3}(\hat{\mathbf{x}}^{(k)}), \quad (36)$$

where $\mathbf{v}^{(k)}$ denotes the predicted log-variance map. The normalized aleatoric uncertainty map is then obtained as

$$\mathbf{AU}^{(k)} = \text{Sigmoid}(\mathbf{v}^{(k)}). \quad (37)$$

Following the heteroscedastic Gaussian uncertainty formulation, the aleatoric uncertainty loss is defined as

$$\mathcal{L}_{\text{au}}^{(k)} = \frac{1}{N} \sum_{j=1}^N \left[\exp(-v_j^{(k)}) \left(x_j^{(k)} - x_j^{\text{gt}} \right)^2 + v_j^{(k)} \right]. \quad (38)$$

Here, N is the total number of image pixels, and k denotes the stage index.

3.4. Loss function

The loss defined below is a supervised training objective optimized with respect to the learnable network parameters and is therefore distinct from the model-driven reconstruction objective. The overall training loss $\mathcal{L}$ consists of three components: the mean squared error (MSE) loss $\mathcal{L}_1$, a one-sided high-frequency regularizer $\mathcal{L}_2$, and a multi-stage uncertainty loss $\mathcal{L}_3$:

$$\mathcal{L} = \mathcal{L}_1 + \rho_1 \mathcal{L}_2 + \rho_2 \mathcal{L}_3, \quad (39)$$

where ρ_1 and ρ_2 are balancing weights, both set to 0.001 in our experiments. The MSE loss $\mathcal{L}_1$ ensures pixel-level fidelity between the final

reconstruction $\mathbf{x}^{(K)}$ and the ground truth $\mathbf{x}^{\text{gt}}$:

$$\mathcal{L}_1 = \frac{1}{N_a N_b} \left\| \mathbf{x}^{(K)} - \mathbf{x}^{\text{gt}} \right\|_2^2, \quad (40)$$

where N_a is the number of training samples and N_b is the number of pixels per image. Let θ_H and $\theta_H^{(K)}$ be the high-frequency subbands of the ground truth and the final reconstruction, respectively. The high-frequency-energy regularizer $\mathcal{L}_2$ is defined as:

$$\mathcal{L}_2 = \max \left(\left\| \theta_H^{(K)} \right\|_2^2 - \left\| \theta_H \right\|_2^2, 0 \right), \quad (41)$$

where $\left\| \cdot \right\|_2^2$ denotes the squared ℓ_2 -norm. This auxiliary term discourages high-frequency energy overshoot relative to the reference. It complements the pixelwise reconstruction loss but does not independently guarantee texture preservation or prevent texture attenuation.

Finally, the uncertainty loss $\mathcal{L}_3$ focuses learning on less certain regions by averaging the epistemic and aleatoric uncertainties into a combined map $U^{(k)}$ for each stage k : $U^{(k)} = 0.5(EU^{(k)} + AU^{(k)})$. The multi-stage uncertainty loss is then given by:

$$\mathcal{L}_3 = \frac{1}{K} \sum_{k=1}^K \left(\left\| U^{(k)} \odot (\mathbf{x}^{(k)} - \mathbf{x}^{\text{gt}}) \right\|_1 + \mathcal{L}_{\text{au}}^{(k)} \right), \quad (42)$$

where the uncertainty map $U^{(k)}$ adaptively reweights pixel-wise reconstruction residuals, encouraging the network to focus more on unreliable regions and difficult structures.

4. Experiments

4.1. Experimental settings

We train on 64×64 patches from 1000 images in COCO train2017 using PyTorch 1.7 and Adam for 100 epochs (batch size 32, learning rate 1×10^{-4}) on an RTX 3090 GPU. The block-based sampling matrix $\mathbf{A}$ is Gaussian-initialized and learnable, following standard learnable-matrix benchmarks in deep learning-based CS [17,51,68,80]. This patch-based sampling follows standard block-wise CS settings and enables fair comparison with existing deep CS methods. Performance is evaluated using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) on five widely used low-level vision datasets: Set11 [29], Set14 [70], General100 [10], Urban100 [26], and OST300 [58]. For all natural-image CS experiments, the model is trained and evaluated on single-channel luminance images. RGB training images and RGB test images from Set14, General100, Urban100, and OST300 are converted to the Y channel. The grayscale Set11 images are used directly. All inputs are normalized to $[0, 1]$ and no averaging over RGB channels is performed. For

Table 2

Average PSNR (dB)/SSIM comparisons with different methods. The best and second-best results are highlighted in **bold** and underlined, respectively.

Dataset	Sampling Matrix	Method	CS Sampling Ratio											
			10%		20%		30%		40%		50%		Avg.	
			PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Set14	Fixed	ISTA-Net [73]	25.85	0.7214	28.91	0.8289	31.37	0.8868	33.48	0.9221	35.60	0.9450	31.04	0.8608
	Learnable	AMP-Net [80]	28.69	0.8171	31.95	0.8933	34.27	0.9293	36.26	0.9505	38.10	0.9647	33.85	0.9110
		COAST [69]	27.41	0.7799	30.71	0.8672	33.10	0.9106	35.12	0.9369	36.93	0.9549	32.65	0.8899
		ISTA-Net++ [68]	26.75	0.7549	30.09	0.8518	32.40	0.8999	34.26	0.9287	35.90	0.9477	31.88	0.8766
		MADUN [49]	27.97	0.7914	31.50	0.8790	34.05	0.9194	36.05	0.9439	37.96	0.9592	33.51	0.8986
		DPUNet [63]	28.49	0.8226	31.61	0.8963	33.95	0.9308	35.93	0.9505	37.66	0.9628	33.53	0.9126
		DPC-DUN [50]	28.02	0.7947	31.37	0.8781	33.92	0.9188	35.91	0.9429	37.83	0.9592	33.41	0.8987
		SODAS-Net [52]	27.54	0.7812	30.32	0.8624	33.62	0.9163	35.72	0.9415	37.61	0.9581	32.96	0.8919
		UFC-Net [57]	29.09	0.8362	–	–	–	–	–	–	–	–	–	–
		MDGF-Net [59]	29.03	0.8356	–	–	34.65	0.9387	<u>36.81</u>	<u>0.9572</u>	<u>38.85</u>	0.9694	–	–
		WTDUN [22]	28.71	0.8235	32.10	0.8944	34.57	0.9308	36.53	0.9507	38.49	0.9653	34.08	0.9129
		PLV-CSNet [24]	<u>29.15</u>	<u>0.8400</u>	<u>32.26</u>	<u>0.9079</u>	<u>34.73</u>	0.9405	36.71	0.9580	38.71	0.9702	<u>34.31</u>	0.9233
		GDU-Net (Ours)	29.64	0.8402	32.88	0.9088	35.40	<u>0.9400</u>	37.32	<u>0.9572</u>	39.39	<u>0.9699</u>	34.93	<u>0.9232</u>
		General100	Fixed	ISTA-Net [73]	28.25	0.8027	31.94	0.8920	34.90	0.9350	37.25	0.9572	39.36	0.9716
Learnable	AMP-Net [80]		31.28	0.8825	35.29	0.9401	38.02	0.9645	40.11	0.9771	41.96	0.9848	37.33	0.9498
	COAST [69]		30.62	0.8589	34.39	0.9238	36.91	0.9520	39.04	0.9681	40.95	0.9783	36.38	0.9362
	ISTA-Net++ [68]		29.37	0.8330	33.14	0.9110	35.69	0.9443	37.74	0.9626	39.44	0.9736	35.08	0.9249
	MADUN [49]		31.13	0.8676	35.15	0.9329	37.86	0.9594	40.08	0.9735	42.18	0.9821	37.28	0.9431
	DPUNet [63]		31.53	0.8896	35.41	0.9430	38.04	0.9650	40.16	0.9767	41.94	0.9836	37.42	0.9516
	DPC-DUN [50]		31.17	0.8716	34.98	0.9321	37.76	0.9590	39.95	0.9731	42.01	0.9820	37.17	0.9436
	SODAS-Net [52]		30.58	0.8602	34.10	0.9224	37.49	0.9576	39.74	0.9722	41.74	0.9813	36.73	0.9387
	UFC-Net [57]		32.31	0.9014	–	–	–	–	–	–	–	–	–	–
	MDGF-Net [59]		<u>32.55</u>	0.9043	–	–	<u>39.08</u>	<u>0.9710</u>	<u>41.32</u>	0.9814	<u>43.43</u>	<u>0.9878</u>	–	–
	WTDUN [22]		31.83	0.8930	35.76	0.9415	38.70	0.9660	40.89	0.9778	43.15	0.9858	<u>38.07</u>	0.9528
	PLV-CSNet [24]		32.40	0.9050	<u>35.81</u>	<u>0.9494</u>	38.20	0.9692	39.54	0.9795	42.57	0.9870	37.70	<u>0.9580</u>
	GDU-Net (Ours)		32.99	<u>0.9044</u>	36.79	0.9513	39.59	0.9711	41.80	0.9814	43.98	0.9881	39.03	0.9593
	Urban100		Fixed	ISTA-Net [73]	23.28	0.7094	26.90	0.8364	29.62	0.8980	31.87	0.9322	33.98	0.9538
Learnable		AMP-Net [80]	25.96	0.8133	29.50	0.8974	32.07	0.9352	34.22	0.9569	36.16	0.9706	31.58	0.9147
		COAST [69]	25.94	0.8038	29.70	0.8940	32.20	0.9317	34.21	0.9528	35.99	0.9665	31.61	0.9098
		ISTA-Net++ [68]	24.78	0.7607	28.55	0.8687	31.08	0.9152	33.10	0.9402	34.86	0.9560	30.47	0.8882
		OPINE-Net [74]	26.56	0.8345	30.07	0.9088	32.64	0.9419	34.66	0.9600	36.64	0.9725	32.11	0.9235
		DPUNet [63]	26.10	0.8226	29.71	0.9027	32.23	0.9378	34.30	0.9573	36.10	0.9693	31.69	0.9179
		DPC-DUN [50]	26.96	0.8361	–	–	<u>33.53</u>	0.9449	<u>35.61</u>	0.9624	37.52	0.9737	–	–
		SODAS-Net [52]	26.22	0.8055	29.51	0.8950	33.15	0.9412	35.27	0.9599	37.14	0.9721	32.26	0.9147
		UFC-Net [57]	<u>27.55</u>	<u>0.8583</u>	–	–	–	–	–	–	–	–	–	–
		MDGF-Net [59]	27.13	0.8501	–	–	33.20	<u>0.9482</u>	35.35	<u>0.9650</u>	37.31	<u>0.9760</u>	–	–
		WTDUN [22]	26.77	0.8402	<u>30.82</u>	0.9142	33.44	0.9459	35.60	0.9629	<u>37.65</u>	0.9748	<u>32.86</u>	0.9276
		PLV-CSNet [24]	26.08	0.8428	30.26	<u>0.9186</u>	32.76	0.9451	34.22	0.9588	36.86	0.9753	32.04	<u>0.9281</u>
		GDU-Net (Ours)	27.86	0.8598	31.61	0.9254	34.28	0.9531	36.51	0.9686	38.54	0.9786	33.76	0.9371

“–” indicates that the result is not reported.

qualitative visualization only, the reconstructed Y channel is combined with the original Cb and Cr channels to generate the displayed color images. The same color-composition procedure is applied to all compared methods. These colorized results are used only for visual illustration and are not involved in the quantitative evaluation; all reported PSNR and SSIM values are computed exclusively on the reconstructed luminance or grayscale channel.

4.2. Comparison with other CS methods

For methods with publicly available official implementations, we retrain and evaluate the models under the unified experimental environment and evaluation protocol described in Section 4.1. For methods without publicly available code, we report quantitative results from the original publications only when the dataset, CS sampling ratio, and sensing protocol are compatible with ours. Results unavailable under a compatible setting are denoted by “–”. Table 2 compares the average PSNR/SSIM on Set14, General100, and Urban100, showing

that GDU-Net consistently achieves competitive or superior performance across different CS sampling ratios. For example, on Urban100 at 10% CS sampling, ours outperforms ISTA-Net [73], AMP-Net [80], COAST [69], ISTA-Net++ [68], OPINE-Net [74], DPUNet [63], DPC-DUN [50], SODAS-Net [52], MDGF-Net [59], and UFC-Net [57] by 4.58 dB, 1.90 dB, 1.92 dB, 3.08 dB, 1.30 dB, 1.76 dB, 0.90 dB, 1.64 dB, 0.73 dB, and 0.31 dB in PSNR, and by 0.1504, 0.0465, 0.0560, 0.0991, 0.0253, 0.0372, 0.0237, 0.0543, 0.0097, and 0.0015 in SSIM. As shown in Table 3, GDU-Net achieves the best or second-best result for every evaluated dataset–metric combination at the very low CS sampling ratios of 1% and 4%. On Urban100 at the 1% CS sampling ratio, GDU-Net obtains 20.14 dB/0.5328: its PSNR is only 0.04 dB below that of SFN-Net [35] (20.18 dB), while its SSIM is the highest among all compared methods. For the remaining settings, GDU-Net achieves the best PSNR and SSIM, demonstrating competitive performance against LTWIST [18], TCS-Net [17], UFC-Net [57], OCTUF [51], and SFN-Net [35].

The visual comparisons in Figs. 4 and 5, and 7 further illustrate the reconstruction advantages of our method. In Fig. 4, our method better

Table 3

Average PSNR (dB)/SSIM comparisons on Urban100 and OST300 at low CS sampling ratios. The best and second-best results are highlighted in **bold** and underlined, respectively.

Dataset	CS ratio	LTWIST [18]	TCS-Net [17]	UFC-Net [57]	OCTUF [51]	SFN-Net [35]	GDU-Net (Ours)
Urban100	1%	19.46/0.4886	19.61/0.4945	19.69/0.5041	19.79/0.5143	20.18/0.5294	20.14/0.5328
	4%	23.00/0.7061	22.93/0.7036	23.37/0.7195	<u>23.76/0.7346</u>	23.48/0.7236	23.92/0.7373
OST300	1%	22.17/0.5105	22.08/0.5088	<u>22.40/0.5225</u>	22.31/0.5253	–/–	22.84/0.5440
	4%	24.84/0.6761	24.69/0.6703	25.00/0.6830	<u>25.21/0.6928</u>	–/–	25.40/0.6974

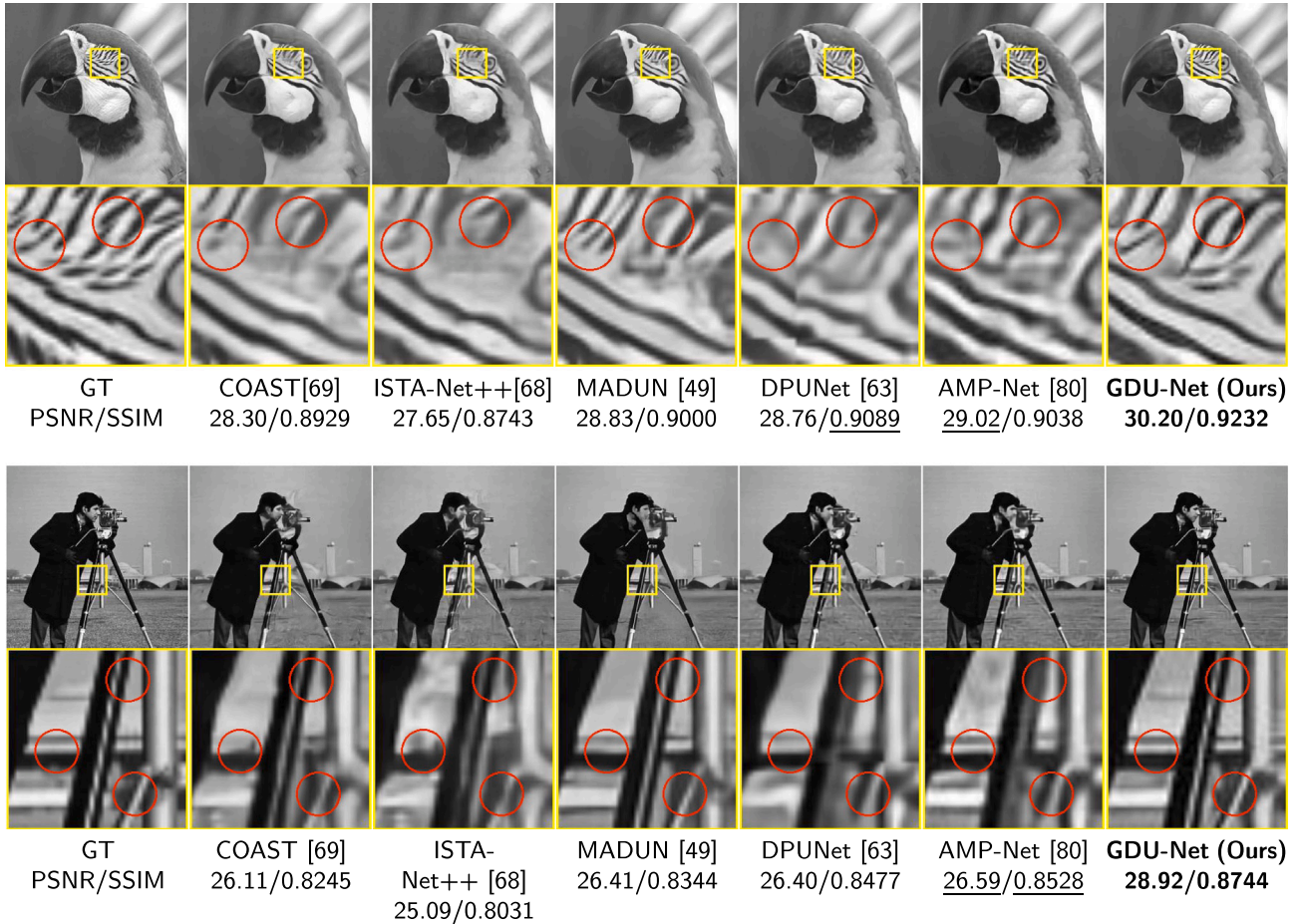


Fig. 4. Visual comparisons on Set11 at 10% CS sampling ratio. The best and second-best results are highlighted in **bold** and underlined, respectively.

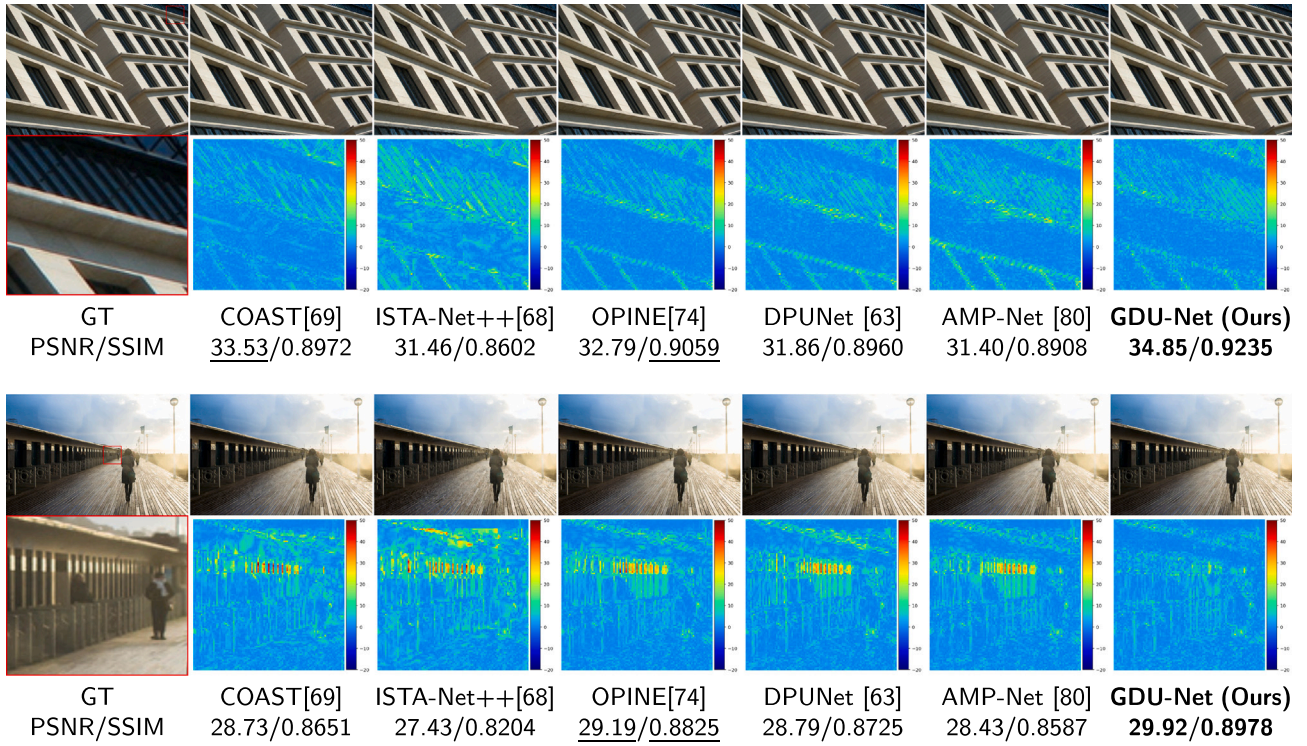


Fig. 5. Visual quality comparisons between the proposed GDU-Net and state-of-the-art CS methods on Urban100 at 10% CS sampling ratio. To better illustrate the differences, we show the residual heatmaps between the reconstructed image patch and the GT patch. The best and second-best results are highlighted in **bold** and underlined, respectively.

preserves intricate structural details and suppresses artifacts. In Fig. 7, it reconstructs high-frequency textures (e.g., fabric folds) with sharpness and geometric consistency, while competing methods show blurring and loss of granularity. As shown in Fig. 5 at a 10% CS sampling ratio, our method shows lower error concentration in residual heatmaps and preserves fine building textures, further illustrating the reconstruction advantages of our method.

4.3. Proof-of-concept generalization to CS-MRI

To evaluate the generalization of our GDU-Net beyond natural images, we conduct a proof-of-concept study on Magnetic Resonance Imaging (MRI). In contrast to block-based image CS, we retrain our network globally on 256×256 brain MRI images, following the standard protocols of ADMM-CSNet [62] and ISTA-Net⁺ [73]. To align with our dual-domain framework, the forward model bridges the image-domain and the Fourier-domain via $\mathbf{A} = \mathbf{P}\mathbf{F}$, where $\mathbf{P}$ is a binary mask and $\mathbf{F}$ denotes the 2D Fourier transform. This model is applied to the whole image, and reconstruction employs this same discrete encoding operator, consistent with all compared baselines [62,73]. While this simulation does not model coil sensitivities, calibration errors, or other hardware-specific acquisition effects, it provides a controlled and reproducible benchmark for algorithmic comparison. The baseline reconstruction comparison uses noiseless simulated measurements. In the separate robustness experiment, independent zero-mean Gaussian noise is added to the real and imaginary components of the undersampled k -space measurements. After reconstruction and magnitude calculation, the resulting image-domain noise exhibits Rician-type magnitude-domain perturbations.

We compare GDU-Net with ADMM-CSNet [62] and ISTA-Net⁺ [73] across CS sampling ratios ranging from 10% to 50%. As shown in Table 4, GDU-Net consistently achieves higher PSNR/SSIM than the compared baselines. Compared with the baseline ISTA-Net⁺, it achieves

the largest PSNR gain of + 0.95 dB at the 10% CS sampling ratio. Visual comparisons in Fig. 6 further demonstrate that our method yields sharper details, cleaner textures, and fewer artifacts. The results in Fig. 9(g) provide a controlled robustness evaluation rather than a complete physical model of MRI acquisition noise.

4.4. Ablation studies and analyses

4.4.1. Impact of dual-domain information

Table 5 compares three ablation settings on Set11 at 10% CS sampling. Case (a) uses image-domain features (global structure), Case (b) uses wavelet-domain features (high-frequency details). The dual-domain Case (c) improves PSNR by 0.16 dB and 0.96 dB over Cases (a) and (b), respectively, due to a more comprehensive representation. Fig. 8 visualizes this complementarity. This performance gain is attributed to our dedicated cross-domain gating, not naive fusion. The three gates (σ_1 , σ_2 , σ_3) are aligned with the gradient descent, denoising, and feedback steps of the unfolded optimizer. This design provides update-specific and bidirectional guidance with explicitly traceable communication sites, distinguishing it from static fusion strategies.

4.4.2. Impact of ATSM and UEM

Table 6 compares three cases: (a) without ATSM, (b) without UEM, and (c) our full model. The full model outperforms (a) and (b) by 0.12 dB/0.0038 and 0.16 dB/0.0044 in PSNR/SSIM, respectively. The best performance is achieved when ATSM and UEM are used together. Combining ATSM and UEM provides a richer representation and avoids the bias or information loss caused by using only one module. Fig. 10(b) shows uncertainty heatmaps at different CS sampling ratios. Brighter regions indicate lower reconstruction reliability. Aleatoric uncertainty (AU) mainly appears in complex textures and high-frequency regions, while epistemic-style cross-domain discrepancy

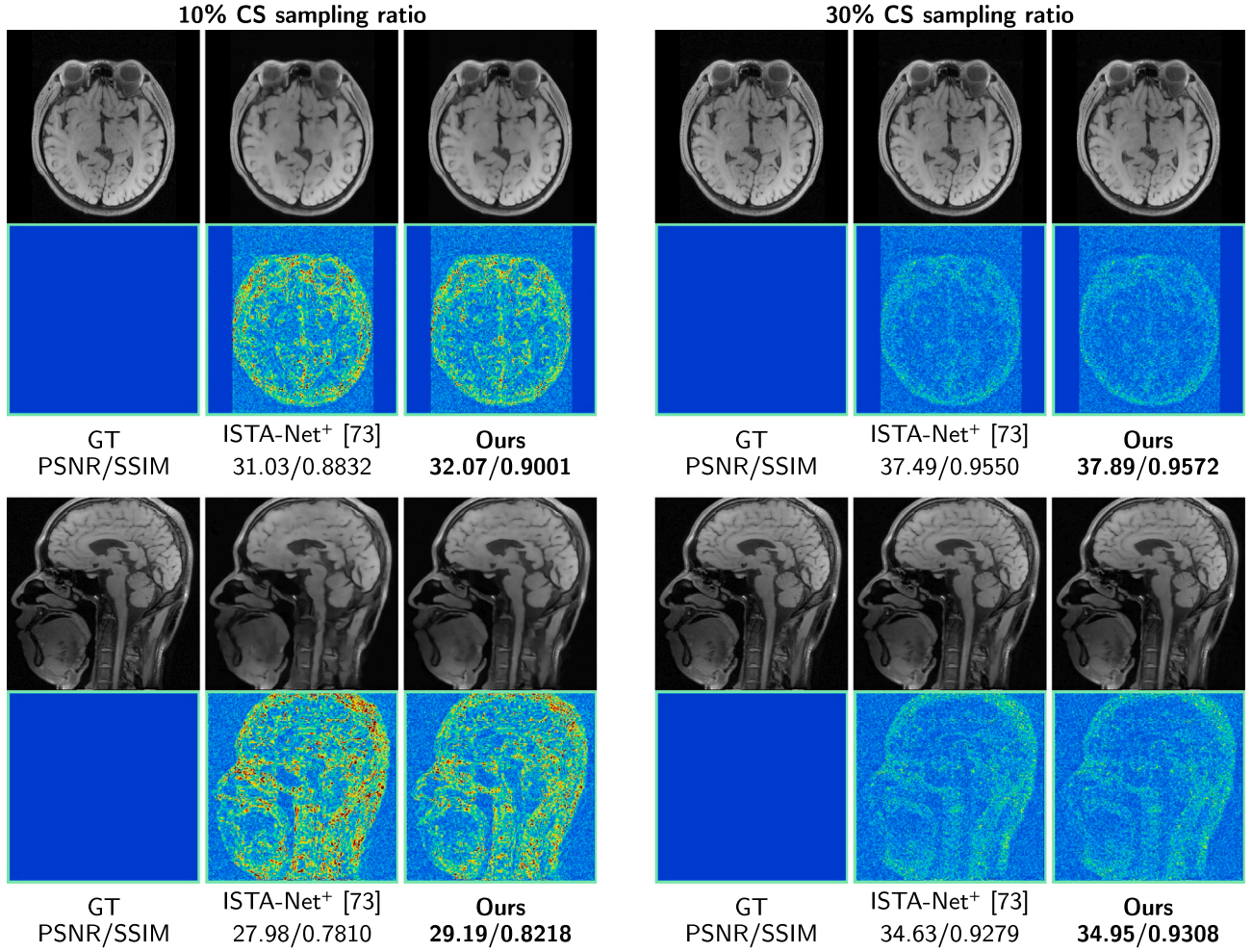


Fig. 6. Visual comparison of CS-MRI reconstructions and their absolute residual maps with respect to the ground truth, evaluated on images “brain test-02” and “brain test-49” under 10% and 30% CS sampling ratios.

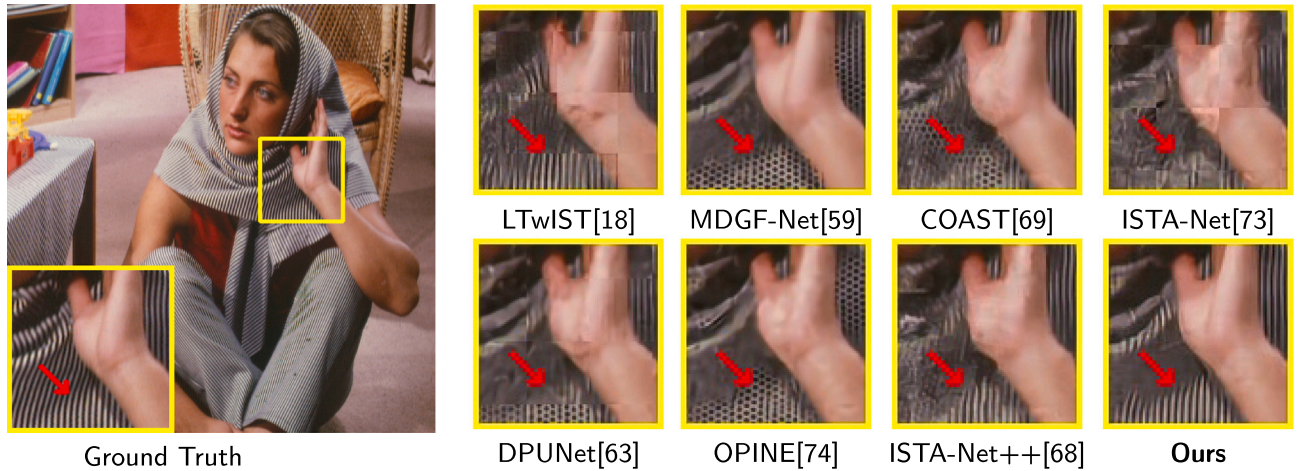


Fig. 7. Visual comparisons on Set14 at 10% CS sampling ratio. Ours achieves superior performance (25.87/0.7943) compared to others: LTwIST (24.69/0.7423), MDGF-Net (25.38/0.7889), COAST (25.33/0.7791), ISTA-Net (24.13/0.7224), DPUNet (25.52/0.7832), OPINE-Net (25.33/0.7820), ISTA-Net++ (24.76/0.7474). Our method better preserves texture details while reducing high-frequency artifacts.

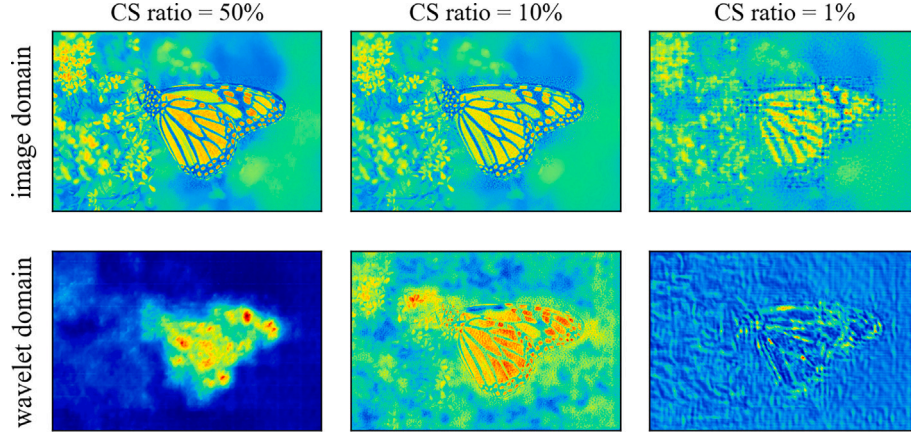
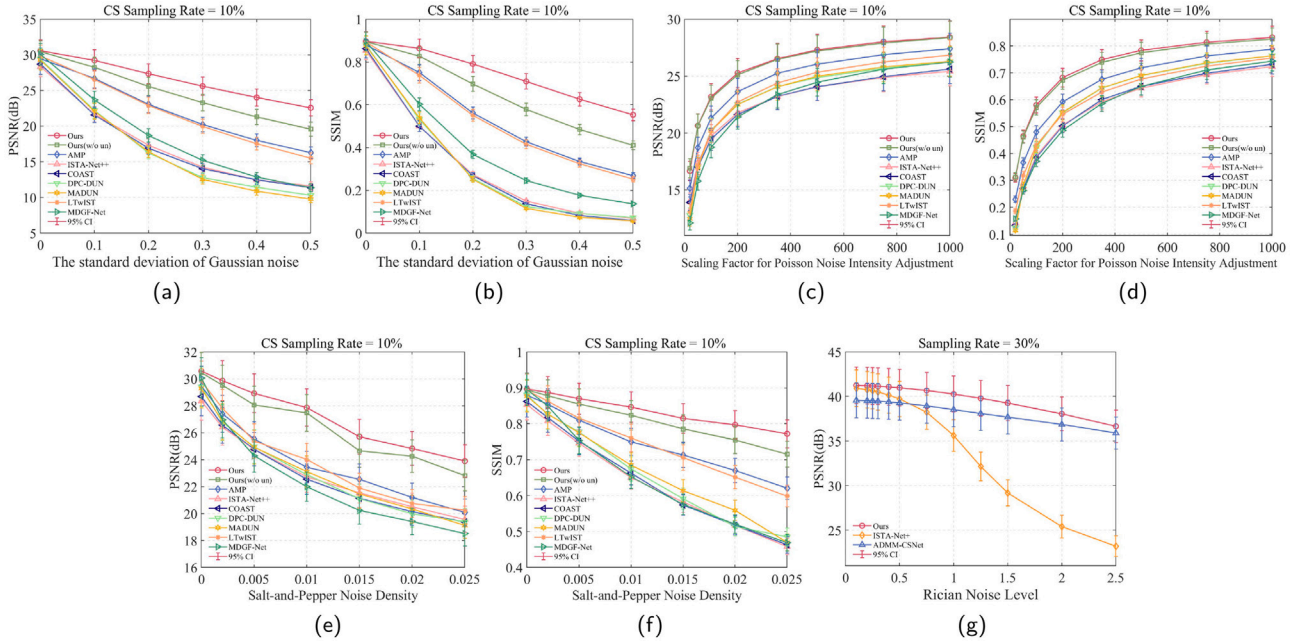
uncertainty (EU) decreases as the CS sampling ratio increases, indicating improved cross-domain reconstruction consistency. Overall, reconstruction reliability varies with both spatial location and CS sampling ratio.

4.4.3. Impact of loss function

To validate loss function settings, we conduct ablation experiments on Urban100 at 10% CS sampling (Table 7). Case (c) outperforms Cases

Table 4Average PSNR (dB)/SSIM comparisons on **CS-MRI** with brain MR images. The best results are highlighted in **bold**.

Method	CS Sampling Ratio									
	10%		20%		30%		40%		50%	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
ADMM-CSNet[62]	34.42	0.8971	38.60	0.9478	40.87	0.9633	42.58	0.9726	44.19	0.9796
ISTA-Net+[73]	34.63	0.9035	38.70	0.9484	40.97	0.9639	42.65	0.9729	44.12	0.9792
GDU-Net (Ours)	35.58	0.9181	39.15	0.9523	41.25	0.9655	42.87	0.9739	44.41	0.9804

**Fig. 8.** The visualization of dual-domain features. The image-domain features highlight global structure information, while the wavelet-domain features focus on texture details. Dual-domain complementarity alleviates the limitations of single-domain representations.**Fig. 9.** Robustness analyses. (a)–(f) Natural-image CS robustness evaluation on Set11 at a 10% CS sampling ratio, reported with 95% confidence intervals: (a)–(b) Gaussian measurement noise with varying standard deviations, (c)–(d) Poisson measurement noise at different levels, and (e)–(f) salt-and-pepper measurement noise with increasing densities. (g) CS-MRI robustness evaluation at a 30% CS sampling ratio, where zero-mean Gaussian noise is added independently to the real and imaginary components of the undersampled k -space measurements. The corresponding magnitude-domain perturbation exhibits Rician characteristics. For simplicity, the horizontal axis is labeled “Rician noise Level”.

(a) and (b) by 0.28 dB and 0.11 dB in PSNR, and by 0.0080 and 0.0033 in SSIM, respectively. A well-chosen loss function accurately measures errors and improves the reported reconstruction metrics, which is essential for improving reconstruction quality under limited data.

4.4.4. Analysis of cross-domain gates

We compare four dual-domain interaction strategies on Urban100 at a 10% CS sampling ratio in Table 8. Cases (a) and (b) use element-wise addition and channel concatenation, respectively. Case (c) replaces

Table 5

Ablation study of single- and dual-domain information at a 10% CS sampling rate. Average PSNR (dB)/SSIM results on Set11 are reported, with the best results highlighted in bold.

Configuration	Domain Information		Set11	
	Wavelet	Image	PSNR	SSIM
(a) Image only		✓	30.41	0.8937
(b) Wavelet only	✓		29.61	0.8798
(c) Dual domain (GDU-Net)	✓	✓	30.57	0.8967

Table 6

Component ablation study on Urban100 at a 10% CS sampling rate. Average PSNR (dB)/SSIM results are reported, with the best results highlighted in bold.

Configuration	Module		Urban100	
	ATSM	UEM	PSNR	SSIM
(a) w/o ATSM		✓	27.74	0.8560
(b) w/o UEM	✓		27.70	0.8554
(c) Full GDU-Net	✓	✓	27.86	0.8598

the three communication gates with independent Channel-Attention Fusion (CAF) modules that receive the same cross-domain inputs at the corresponding communication sites. CAF represents one controlled channel-attention baseline rather than all possible cross-domain attention mechanisms. Case (d) uses the proposed gated communication. The proposed design improves PSNR by 0.15 dB, 0.11 dB, and 0.16 dB over Addition, Concatenation, and CAF, respectively, and also achieves the highest SSIM. These results support the effectiveness of the proposed gated parameterization at the prescribed communication sites under the tested setting.

Table 9 further verifies the contribution of each cross-domain gate in DDC. Removing any gate leads to performance degradation, indicating that the three gates play complementary roles in the unfolding process. In particular, removing the dynamic interactive gate σ_2 causes the largest drop on both Set11 and Urban100, suggesting that adaptive cross-domain interaction during the denoising step is

Table 7

Ablation study of the auxiliary loss components on Urban100 at a 10% CS sampling rate. All configurations include the MSE loss. Average PSNR (dB)/SSIM results are reported, with the best results highlighted in bold.

Configuration	Auxiliary Loss		Urban100	
	Wavelet	Uncertainty	PSNR	SSIM
(a) MSE + uncertainty		✓	27.58	0.8518
(b) MSE + wavelet	✓		27.75	0.8565
(c) Full objective	✓	✓	27.86	0.8598

Table 8

Comparison of different dual-domain interaction strategies on Urban100 at a 10% CS sampling rate. Average PSNR (dB)/SSIM results are reported, with the best results highlighted in bold.

Case	Interaction Strategy	PSNR	SSIM
(a)	Addition	27.71	0.8539
(b)	Concatenation	27.75	0.8577
(c)	Channel-Attention Fusion	27.70	0.8555
(d)	Gated interaction (proposed)	27.86	0.8598

Table 9

Ablation study of the three DDC gates on Set11 and Urban100 at a 10% CS sampling rate. Average PSNR (dB)/SSIM results are reported, with the best results highlighted in bold.

Configuration	DDC Gate			Dataset	
	σ_1	σ_2	σ_3	Set11	Urban100
(a) w/o σ_1		✓	✓	30.51/0.8956	27.80/0.8568
(b) w/o σ_2	✓		✓	30.44/0.8947	27.70/0.8552
(c) w/o σ_3	✓	✓		30.55/0.8955	27.76/0.8559
(d) Full DDC	✓	✓	✓	30.57/0.8967	27.86/0.8598

especially important for reconstruction quality. The full model consistently achieves the best performance, demonstrating that prior injection, dynamic interaction, and state feedback jointly contribute to effective dual-domain communication.

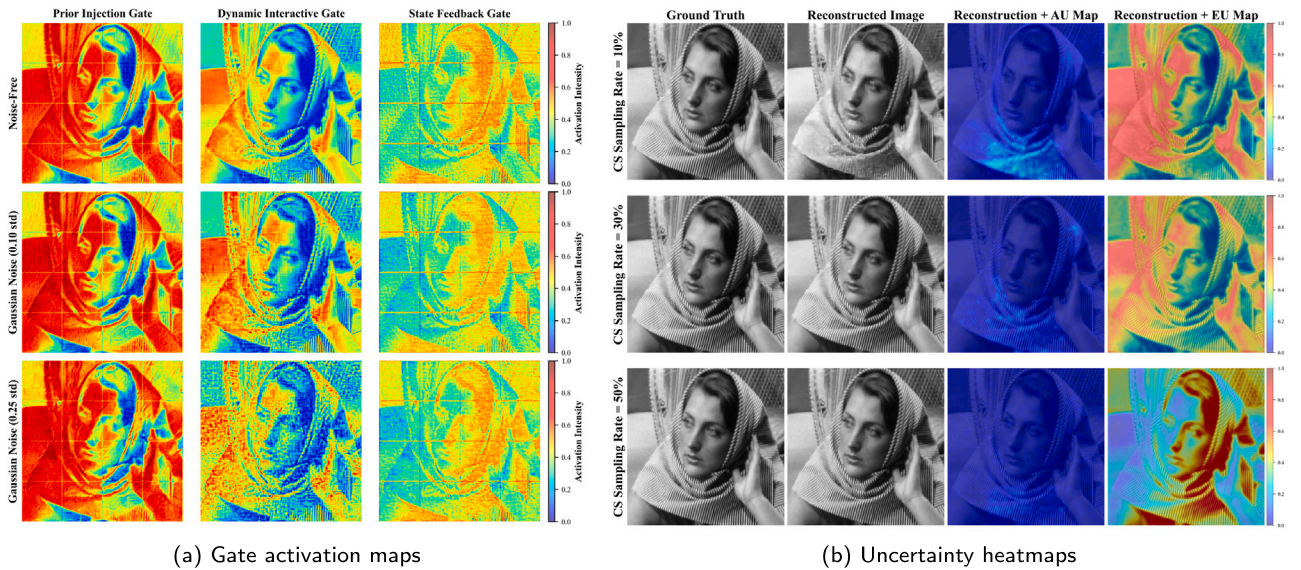


Fig. 10. Visualization of gate activations and reconstruction uncertainty. (a) Gate activation maps under different noise levels, showing spatially varying and condition-dependent activations of the three communication gates. (b) Reconstruction and uncertainty heatmaps under different CS sampling ratios, where brighter/warmer regions indicate higher estimated uncertainty and thus lower reconstruction reliability.

Table 10

Comparison of computational complexity, inference efficiency, and reconstruction performance at the 10% CS sampling ratio. All profiling measurements are conducted using a unified fixed 256×256 single-channel input with a batch size of one, whereas PSNR/SSIM is averaged over all 11 Set11 images at their native resolutions.

Method	Params. (M)	FLOPs (G)	Time (s)	Peak memory (MB)	PSNR (dB)/SSIM
ISTA-Net [73]	0.76	35.07	0.07	184.48	26.26/0.7961
OPINE-Net [74]	0.80	35.98	0.08	124.48	29.81/0.8904
AMP-Net [80]	0.22	25.84	0.07	30.94	29.27/0.8768
MADUN [49]	2.60	176.49	0.12	232.96	29.29/0.8768
DPC-DUN [50]	1.05	70.06	0.09	116.76	29.40/0.8798
OCTUF [51]	0.29	18.85	0.10	118.94	30.70/0.9030
LTWIST [18]	22.05	110.74	0.26	253.86	29.83/0.8930
NesTD-Net [16]	5.07	343.43	0.22	998.77	28.23/0.8561
UFC-Net [57]	1.54	112.77	0.24	204.54	30.14/0.8960
GDU-Net (Ours)	1.64	77.53	0.26	226.96	30.57/0.8967

Table 11

Comparison of parameters, FLOPs, inference time, and reconstruction performance across different numbers of unfolding stages at a 10% sampling rate. PSNR and SSIM are averaged over all 11 Set11 images at their native resolutions. FLOPs and inference time are measured using 256×256 single-channel inputs. The default setting is highlighted in bold.

Metric	$K = 3$	$K = 6$	$K = 9$	$K = 10$	$K = 11$
PSNR (dB)	29.71	30.29	30.57	30.42	30.64
SSIM	0.8825	0.8912	0.8967	0.8940	0.8968
Parameters (M)	0.54	1.09	1.64	1.83	2.02
FLOPs (G)	25.36	51.16	77.53	86.47	95.49
Inference time (s)	0.13	0.20	0.26	0.29	0.31

4.4.5. Robustness to noise

We evaluate robustness by adding different types of noise to CS measurements on Set11 at a 10% CS sampling ratio. Across Gaussian, Poisson, and salt-and-pepper measurement noise, GDU-Net generally achieves the highest or competitive mean PSNR/SSIM and exhibits a slower degradation trend than the compared methods, as shown in Fig. 9(a)–f. For the CS-MRI robustness test, independent zero-mean Gaussian noise is added to the real and imaginary components of the undersampled k -space measurements, and the same noise realizations and noise levels are used for all compared methods. The resulting magnitude reconstructions exhibit Rician-type perturbations. As shown in Fig. 9(g), GDU-Net achieves the highest mean PSNR among the compared methods at all tested noise levels.

4.4.6. Complexity analysis

Table 10 reports reconstruction accuracy and computational complexity. PSNR and SSIM are averaged over all 11 Set11 images at

their native resolutions under the 10% CS sampling ratio. Parameter counts are reported for each model, whereas FLOPs, runtime, and peak allocated GPU memory are measured using a unified 256×256 single-channel input with a batch size of one. The same profiling wrapper, input configuration, and FLOP-counting convention are applied to all compared methods. Accordingly, the reported FLOPs are used only for relative computational-complexity comparison within this unified protocol. OCTUF is substantially lighter than GDU-Net and also obtains slightly higher Set11 PSNR/SSIM. GDU-Net achieves 30.57 dB/0.8967 on Set11 with 1.64 M parameters and 77.53 G reported FLOPs. This comparison characterizes the accuracy–complexity trade-off of the proposed dual-domain and uncertainty-aware design.

4.4.7. Discussion on the number of stages

Table 11 summarizes the reconstruction accuracy and computational cost under different numbers of unfolding stages. PSNR and SSIM are computed on all 11 Set11 images at their native resolutions and then averaged with equal weight, without cropping or resizing. FLOPs are measured using a fixed 256×256 single-channel input, whereas inference time is averaged over the nine native 256×256 inputs. Compared with $K = 9$, $K = 11$ increases PSNR/SSIM by 0.07 dB/0.0001, while increasing the parameter count from 1.64 M to 2.02 M, the reported FLOPs from 77.53 G to 95.49 G, and inference time from 0.26 s to 0.31 s. We therefore select $K = 9$ as the default configuration because it provides a favorable reconstruction-quality–efficiency trade-off. Each depth configuration is trained independently under the same training protocol; therefore, small non-monotonic variations across different values of K may arise from optimization stochasticity. To complement the quantitative accuracy–efficiency comparison, Fig. 11 presents the final reconstruction results produced by independently trained GDU-Net models with $K = 6$, $K = 9$, $K = 10$, and $K = 11$ unfolding stages. The

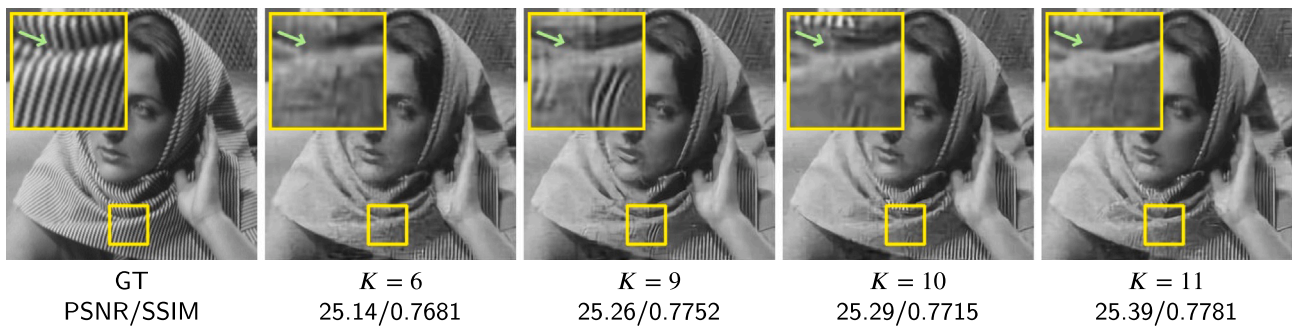


Fig. 11. Visual comparison with different unfolding stages at a 10% CS sampling rate. The stripe transitions become clearer from $K = 6$ to $K = 9$, whereas the additional visible changes from $K = 9$ to $K = 10, 11$ are comparatively limited in this illustrative example. The values below the reconstructed images denote PSNR (dB)/SSIM.

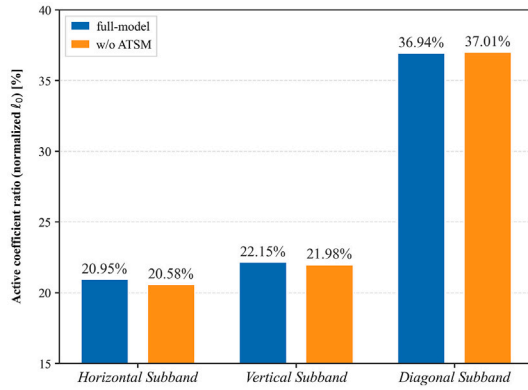


Fig. 12. Comparison of normalized ℓ_0 ratios across high-frequency wavelet subbands at a 10% CS sampling ratio on the Urban100 dataset.

same texture region, crop, and intensity range are used for all configurations. Compared with the $K = 6$ model, the $K = 9$ model reconstructs clearer and more complete stripe transitions.

4.4.8. Analysis of ATSM on high-frequency wavelet coefficient activation

To further analyze the effect of ATSM in the wavelet domain, we compare GDU-Net with a variant without ATSM on the Urban100 dataset under a 10% CS sampling ratio. We apply one-level wavelet decomposition to the reconstructed images and compute the normalized ℓ_0 ratio for each high-frequency subband, including the horizontal, vertical, and diagonal subbands. Here, the normalized ℓ_0 ratio reflects the proportion of active wavelet coefficients, rather than a strict sparsity score. As shown in Fig. 12, GDU-Net obtains normalized ℓ_0 ratios of 20.95%, 22.15%, and 36.94% in the horizontal, vertical, and diagonal subbands, respectively, while the variant without ATSM obtains 20.58%, 21.98%, and 37.01%. Compared with the variant without ATSM, GDU-Net slightly increases the active coefficient ratios in the horizontal and vertical subbands by 0.37 and 0.17 percentage points, with a marginal decrease of 0.07 percentage points in the diagonal subband. These results suggest that ATSM adaptively redistributes high-frequency wavelet responses across subbands, rather than simply increasing or decreasing coefficient sparsity uniformly. This behavior is consistent with its role in learning data-dependent tree-structured modulation for preserving directional details.

These ablation results do not aim to show that each component independently yields a large gain; rather, they verify that optimization-aligned communication, wavelet-tree modulation, and uncertainty-aware reliability control consistently contribute to the final reconstruction process. This consistent trend supports the design rationale of GDU-Net, where different components play complementary roles along the dual-domain unfolding trajectory.

5. Conclusion

This paper presents GDU-Net, a dual-domain unfolding network for image CS reconstruction. The coupled reconstruction objective identifies the bidirectional dependencies between the explicit image and wavelet-domain states, while three learned gates instantiate wavelet-prior-conditioned image updating, dual-domain denoiser conditioning, and image-to-wavelet feedback at prescribed positions of each unfolded stage. The elementary gating operations are standard; the proposed design focuses on their update-specific communication roles. ATSM provides learnable tree-structured wavelet modulation, and UEM predicts spatial reliability information for uncertainty-aware refinement. Experiments demonstrate competitive reconstruction performance and robustness under the tested simulated sampling and noise conditions. The current study does not include compressive measurements acquired

from a physical imaging system. Extending the method to real acquisition settings with hardware noise and system-specific degradation remains an important direction for future work.

CRediT authorship contribution statement

Kai Han: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Jin Wang:** Writing – review & editing, Supervision, Funding acquisition. **HanQin Cai:** Writing – review & editing. **Yunhui Shi:** Writing – review & editing, Supervision, Funding acquisition. **Nam Ling:** Writing – review & editing. **Baocai Yin:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

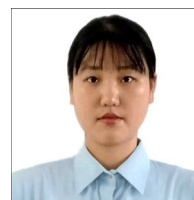
Data will be made available on request.

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